

Ahmed Muhammad

Machine Learning Engineer

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Summary

Machine Learning Engineer with experience building and deploying end-to-end ML systems, from data analysis and processing to model integration and production deployment. Strong in Python-based data pipelines (batch, streaming, async), classical ML, and MLOps practices. Focused on designing reusable, extensible systems and delivering solutions that meet performance and business requirements.

Work Experience

Data Scientist — Singularity Labs

June 2024 – July 2025

AI consultancy delivering data-driven solutions for enterprise clients. **Client:** Mobica

• Mobica Find – AI-Powered Product Search System:

- Delivered a full end-to-end ML solution—covering data collection, analysis, annotation, feature engineering, model integration, and production deployment—reducing product identification time from days to under 3 ms per query.
- Conducted extensive data analysis and processing on product catalogs, standardizing image and text data pipelines to ensure high-quality embedding generation using ALIGN multimodal models.
- Designed a reusable, extensible similarity search architecture backed by Qdrant vector database, supporting incremental updates and cosine-similarity ranking—ensuring the solution was flexible enough for future product lines.
- Deployed a production-ready, API-driven retrieval service with monitoring in place to track query latency and result quality, meeting key performance indicators for accuracy and response time.
- Collaborated cross-functionally with backend and DevOps teams to integrate the ML system into the client's existing infrastructure, promoting shared tooling and common approaches.

STEM Instructor — iSchool – Remote

July 2025 – Sep 2025

- Delivered interactive lessons on programming and foundational CS concepts; strengthened communication and the ability to convey complex technical ideas clearly.

Projects

MisrLeX – Egyptian Legal RAG Platform — Portfolio: ahmed-portfolio-4gxs.vercel.app/

- Designed and built an end-to-end RAG platform for legal document QA, handling the full ML delivery cycle from data ingestion and processing to deployment and monitoring.
- Architected a scalable backend using FastAPI and PostgreSQL (pgvector), implementing semantic search with thorough data analysis and preparation of legal corpora.
- Implemented a Provider Factory design pattern to modularize LLM integration (OpenAI, Cohere, HuggingFace), ensuring a reusable, flexible, and extensible architecture.
- Established MLOps and DevOps workflows: built GitHub Actions CI/CD pipelines, containerized deployment via Docker Compose, and integrated Prometheus and Grafana for real-time system monitoring and alerting.
- Documented technical solutions end-to-end, including API specifications, system architecture, and operational runbooks for ongoing support.

Technical Skills

- **Languages:** Python, SQL, C++
- **Backend & APIs:** FastAPI, SQLAlchemy
- **Data Processing & Pipelines:** Pandas, NumPy, asyncio (batch, streaming, asynchronous workflows)
- **Machine Learning:** Scikit-learn, XGBoost, Optuna (feature engineering, model evaluation)
- **Deep Learning:** PyTorch, TensorFlow (CNNs, RNNs, Transformers)
- **Generative AI:** RAG, LangChain, LangGraph, LlamaIndex
- **MLOps & Deployment:** MLflow, Docker, Kubernetes (familiar), GitHub Actions (CI/CD)
- **Monitoring:** Prometheus, Grafana
- **Databases:** PostgreSQL (pgvector), Qdrant, MongoDB, MySQL

Education

National Telecommunication Institute (NTI)

Cairo, Egypt

Computer Vision Track

April 2026 – Present

- Pursuing intensive training in Computer Vision, deepening expertise in image processing, object detection, and deep learning for vision applications.

Nile University

Cairo, Egypt

Bachelor's Degree in Computer Science

July 2024

Graduation Project: Medical Chatbot Assistant

- Fine-tuned Llama LLM on the MAQA dataset (430k+ Arabic medical Q&A pairs), managing the full model lifecycle: data analysis, annotation review, training, evaluation, and deployment.
- Architected a Node.js backend to manage API requests and orchestrate LLM inference, enabling real-time symptom processing with documented API endpoints.
- Delivered a specialized NLP solution navigating complex Arabic medical terminology, demonstrating strong problem-solving and data-driven reasoning.

Certifications

- Hugging Face Agents Course – Hugging Face, Aug 2025
- Practical Multi AI Agents and Advanced Use Cases with CrewAI – DeepLearning.AI, Jul 2025
- Building Code Agents with Hugging Face smolagents – DeepLearning.AI, Jun 2025
- Fundamentals of Agents – Hugging Face, Apr 2025
- Microsoft Power BI Desktop for Business Intelligence – Maven Analytics, Nov 2024
- Thinking Like an Analyst – Maven Analytics, Oct 2024
- Machine Learning Scientist with Python – DataCamp, Jan 2024
- SQL for Data Science – Coursera, Sep 2023
- IBM Data Science – IBM, Sep 2023
- Introducing Generative AI with AWS – Udacity

Languages

- **Arabic:** Native
- **English:** B2